Class 11: Large Systems and Statistical Mechanics

Our discussions so far have enabled us to recognize that we are dealing with a system that consists of a large collection of particles in the form of free electrons in a solid. We wish to predict the properties of such a system as a whole without having to know the details of every single particle that goes to make up the system – a feat that we can accomplish using an approach that is referred to as 'Statistical Mechanics'.

Before we examine and use the ideas of statistical mechanics, there is a detail which we shall now examine, that we should be alert to since it will be central to much of our subsequent discussion. The need for this digression will become clear as we return to the topic of statistical mechanics a little later in this class. Systems consisting of a large collection of particles are encountered in books and classes associated with Chemistry, Chemical Engineering, Metallurgy, and Physics. While the central ideas are essentially the same, there is a subtle difference in how the topics are handled in these different disciplines. The subtle difference in approach results in different aspects of the results being highlighted in the various disciplines. In general, the difference is as follows:

- In Chemistry and Chemical Engineering based discussions, systems are examined under constant Temperature and Pressure.

- In Metallurgy based discussions also systems are examined under constant Temperature and Pressure, with the additional constraint that the pressure is usually 1 atmosphere.

- In Physics based discussions, systems are examined under constant Temperature and Volume.

We must immediately note that there is an element of generalization in the above statements, and exceptions do exist. However, in general they are true and many standard texts will use the approaches indicated above. Since textbooks are often aimed at students of a specific stream, this variation in approach is often not elaborated on but merely assumed as necessary for that specific stream and left at that. However it is informative to understand the reasons behind, and significance of, such a variation in approach. We will now examine this variation.

In general, to study any system, certain parameters have to be held constant, within the framework of which the system is examined. Usually it is sufficient if two parameters are held constant, or controlled. The state of the system is then fixed and predictions can be made about the system.

In Chemistry as well as in Chemical Engineering liquids and gases are typically examined under experimentally controlled circumstances such as during the refining of

oil. From the perspective of experimental control, temperature and pressure are much easier to control and manipulate. With a liquid, for example, routinely changing its volume in a controlled or predetermined manner,cannot be accomplished easily experimentally. Experimental ease is therefore the reason temperature and pressure are chosen as the variables of choice. Phase diagrams in Chemical Engineering text books therefore typically have pressure and temperature as two of the axis.

In Metallurgical Engineering too, emphasis is placed on experimentally controllable parameters as it pertains to ore extraction and steel production. However, due to the predominant need to handle solids and liquid melts, and large quantities of the materials involved, much of the processing is carried out under 1 atmosphere pressure. Often the experimental work will be done in open air atmosphere itself. Phase diagrams in Metallurgical Engineering text books often plot composition versus temperature, with the understanding that the diagram has been drawn under 1 atmosphere of pressure.

In Physics, to carry out theoretical studies, it is necessary to know the energy levels present within the system and then to look at how these energy levels are occupied by the particles in the system. As we will see in a later class, it turns out that if the volume of the system is fixed, the allowed energy levels of the system get fixed. It is for this specific reason that Physics based approach to discussing systems consisting of large particles use constant volume and temperature as the setting within which the system is examined.

While the above discussion has highlighted the differences between the approaches used, this is not, by any means, intended to suggest that the results obtained using these different approaches cannot be compared. If a surface is used to plot all possible allowed states of the system, the different approaches mentioned above merely refer to different points on this same surface and therefore are not inconsistent with each other. Therefore the different approaches merely suggest different perspectives on the same system. There is no difference in a fundamental sense. It is similar to looking at the same mountain from different locations.

In the present text book we wish to examine the Physics behind material properties, and therefore we will adopt the approach of constant Temperature and constant Volume for our calculations. As we proceed, this assumption will remain in the background but will impact our discussions accordingly. For the present we will merely accept the impact of constant volume on the system. In a later class we will explore how and why constant volume enables us to fix the energy levels present in the system, and therefore justify what we will merely accept for now.

The systems we are interested in are very large in that they contain 10^{28} particles per cubic meter. Our discussions will revolve around energy of such a system and how the energy of the system behaves, for example, how the energy goes up or down with changes in temperature as indicated by its specific heat.

The requirement that a system is at a fixed temperature is associated with the necessity that the constituents of the material have, on average, an energy corresponding to that

temperature. Given the energy of a large system, it is not possible for us to know the energy of each of the 10^{28} atoms or free electrons that go to make up the system. It is not physically possible for us to determine all of the details of each of the particles and then arrive at an average for subsequent use. We need an approach that enables us to make statements about the system as a whole without knowing the all of the details of every single particle in the system. As indicated earlier, such an approach is referred to as statistical mechanics.

We will now look at a few very small systems and see if the understanding we can generate from such systems will give us some indication of how to handle large systems.

Let us also introduce a couple of terms that we are more likely to encounter in a book on Thermodynamics – 'Macrostate' and 'Microstate'. Specifying the macroscopic details of the system such as stating that it is constant temperature, constant volume, and that its total energy is fixed at some specified value – these details together are referred to as the macrostate of the system. Specifying additional microscopic detail of the system such as how many particles are occupying which energy level, consistent with the overall energy of the system, amounts to specifying the microstate of the system. Since we can vary the numbers of particles occupying the various energy levels while still keeping the total energy constant, we can have several microstates corresponding to the same macrostate.

Let us now look at three examples of systems having a fixed macrostate, and see what generalization can be made about their microstates. In all of the examples, let us assume that the system is at a constant temperature, has a constant volume and that its total energy is fixed at a value 3E. Since the system has a fixed volume, we will assume that it has fixed allowed energy levels – we will establish this in a later class, but just assume it for now. Let the allowed energy levels in the system be 0, E, 2E, and 3E. Let us look at examples where the system consists of either 2 particles, or 3 particles or 4 particles. Each arrangement of particles in the allowed energy levels that makes the overall system consistent with the macrostate of 3E, is an allowed microstate of the system.

(The calculations below assume identical but distinguishable particles – an aspect that will explore in greater detail in a subsequent class)

System consisting of 2 particles:

The total energy of 3E can be accomplished in two ways (or using two microstates) in this system. In one microstate, which we shall call microstate-1, there is one particle at the energy level 3E and the other particle is at the energy level 0. In microstate-2 there is one particle each at the energy levels E and 2E. An examination of the system will show that there are no other ways to accomplish the same macrostate.

Microstate-1 can therefore be accomplished in $\frac{2!}{1!*1!}$ ways = 2 ways

and Microstate-2 can be accomplished in $\frac{2!}{1!*1!}$ ways = 2 ways

The table below summarizes this information.

	Number of particles at each energy level				Number of ways to accomplish this Microstate
Energy level →	0	E	2E	3E	
Microstate 1	1	0	0	1	2
Microstate 2	0	1	1	0	2

Since there is no reason why any specific arrangement should be preferred over the rest, a random snapshot of the two particle system will show the particles to be in either of the two microstates with equal probability.

System consisting of 3 particles:

The total energy of 3E can be accomplished through three different microstates. In microstate-1, there is one particle at the energy level 3E and the other two particles are at the energy level 0. In microstate-2 there is one particle each at the energy levels 0, E and 2E, and in microstate-3 all three particles are at the energy level E. An examination of the system will show that there are no other ways to accomplish the same macrostate.

Microstate-1 can be accomplished in $\frac{3!}{2!*1!}$ ways = 3 ways

Microstate-2 can be accomplished in $\frac{3!}{1!*1!*1!}$ ways = 6 ways

Microstate-3 can be accomplished in $\frac{3!}{3!}$ ways = 1 way.

The table below summarizes this information.

	Number of particles at each energy level				Number of ways to accomplish this Microstate
Energy level →	0	E	2E	3E	
Microstate 1	2	0	0	1	3
Microstate 2	1	1	1	0	6
Microstate 3	0	3	0	0	1

The primary information that is of importance from the table is that out of 10 possible microstates to accomplish the same macro state of total energy of 3u, Microstate 2, occurs 6 times. Since there is no reason to assume that any one microstate is preferred over the others, if one takes a snapshot of the system, the probability of observing any given microstate is simply proportional to the number of microstates of that type. In the above example, Microstate-1 is likely to be observed 30% of the time, while Microstate-2 is likely to be observed 60% of the time, and Microstate-3 is likely to be observed 10% of the time. Therefore, in the present example, Microstate-2 is the 'most probable state', and occurs 60% of the time. In other words, the most probable state is more probable than all the other states combined. In this case, one and half times more probable than all of the other states combined, and is twice as probable as the next most probable state.

System consisting of 4 particles:

The total energy of 3E can be accomplished through three different microstates. In microstate-1, there is one particle at the energy level 3E and the other three particles are at the energy level 0. In microstate-2 there is one particle each at the energy levels E and 2E, and two particles at the energy level 0. And in microstate-3 three particles are at the energy level E and one particle is at the energy level 0. An examination of the system will show that there are no other ways to accomplish the same macrostate.

Microstate-1 can be accomplished in $\frac{4!}{3!*1!}$ ways = 4 ways

Microstate-2 can be accomplished in $\frac{4!}{1!*1!*2!}$ ways = 12 ways

Microstate-3 can be accomplished in $\frac{4!}{3!*1!}$ ways = 4 ways.

The table below summarizes this information.

	Number of particles at each energy level				Number of ways to accomplish this Microstate
Energy level →	0	E	2E	3E	
Microstate 1	3	0	0	1	4
Microstate 2	2	1	1	0	12
Microstate 3	1	3	0	0	4

The data in the table indicates that out of 20 possible microstates to accomplish the same macrostate of total energy of 3E, microstate-2, occurs 12 times. Since there is no reason to assume that any one microstate is preferred over the others, if one takes a snapshot of the system, the probability of observing any given microstate is simply proportional to the number of microstates of that type. In the above example, microstate-1 is likely to be observed 20% of the time, while microstate-2 is likely to be observed 60% of the time, and microstate-3 is likely to be observed 20% of the time. Therefore, in the present example, microstate-2 is the 'most probable state', and is more probable than all of the other microstates combined, and is three times more probable than next most probable state.

From the examples above we can make the following observations:

1) As the number of particles increase the microstates are not equally probable. One microstate becomes more probable. In the examples above, as number of particles increases from 2 to 3 and the to 4, the ratio of the occurrence of the most probable state to the next most probable state changes from 1:1, to 2:1, to 3:1. This general trend will hold as we keep increasing the number of particles – the most probable state will keep becoming more probable than the other states in the system combined.
2) The systems we are interested in have 10^{28} particles. The trend of the most probable state becoming more probable becomes significantly magnified in this case. In this type of a large system, the most probable state overwhelms the

probability of occurrence of all other states combined, i.e. the probability of occurrence of the most probable microstate is nearly 100%. Therefore, in large systems, the most probable microstate is effectively the equilibrium state of the system, the state in which it will be found almost all of the time, and hence the only microstate worth further consideration.

Statistical mechanics takes advantage of this situation, and while the mathematics to calculate all possible states may be quite laborious, statistical mechanics ignores all of the other states and focuses on identifying only the most probable microstate, which is significantly easier to do.

We can further understand the generalization with respect to large system as follows. If there are N_f free electrons per unit volume, the general form of the term giving the number of ways in which a microstate can be accomplished will be of the form:

$$\frac{N_f!}{n_1!\ n_2!\ n_3!\ \ldots}$$

Where n_1, n_2, n_3, etc are the numbers of particles at the various energy levels within the system. At very large numbers, the removal of even a single term from the denominator will change the value of the expression above by several orders of magnitude. Therefore the most probable state of the system will occur many orders of magnitude more often than the next most probable state, and more often than all of the other states combined. Statistical mechanics acknowledges that the system can posses other microstates, other than the most probable state, but argues that the chance that the system will be found in the other states is infinitesimally small and hence can be ignored. Statistical mechanics provides us with the tools to understand the most probable state. It enables us to determine properties of large systems without knowing the details of every single particle in the system. It will enable us to see where the error in C_V^e and $\langle v^2 \rangle$ came from. Therefore statistical mechanics is the tool we will use to look at these large systems.